

Experiment No-1

Introduction to MATLAB

AIM: In this lab, we explore the fundamental aspects of MATLAB, a high-performance computing environment widely used in engineering and scientific research. The objectives include gaining familiarity with the MATLAB interface, understanding basic syntax, and executing simple commands.

Software Required: Matlab software.

Theory:

Introduction

MATLAB, developed by MathWorks, is a powerful tool for numerical computing, data analysis, and visualization. Its interactive environment and extensive library of functions make it a valuable asset in various fields. This lab aims to introduce students to MATLAB's basics, preparing them for more advanced applications in subsequent coursework.

A minimum MATLAB session

The goal of this *minimum* session (also called *starting* and *exiting* sessions) is to learn the first steps:

- How to log on
- Invoke MATLAB
- Do a few simple calculations
- How to quit MATLAB

Starting MATLAB

After logging into your account, you can enter MATLAB by double-clicking on the MATLAB shortcut *icon* (MATLAB 7.0.4) on your Windows desktop. When you start MATLAB, a special window called the MATLAB desktop appears. The desktop is a window that contains *other* windows. The major tools within or accessible from the desktop are:

- The COMMAND WINDOW
- TheCOMMANDHISTORY
- The WORKSPACE
- The CURRENT DIRECTORY
- The HELP BROWSER
- The START button

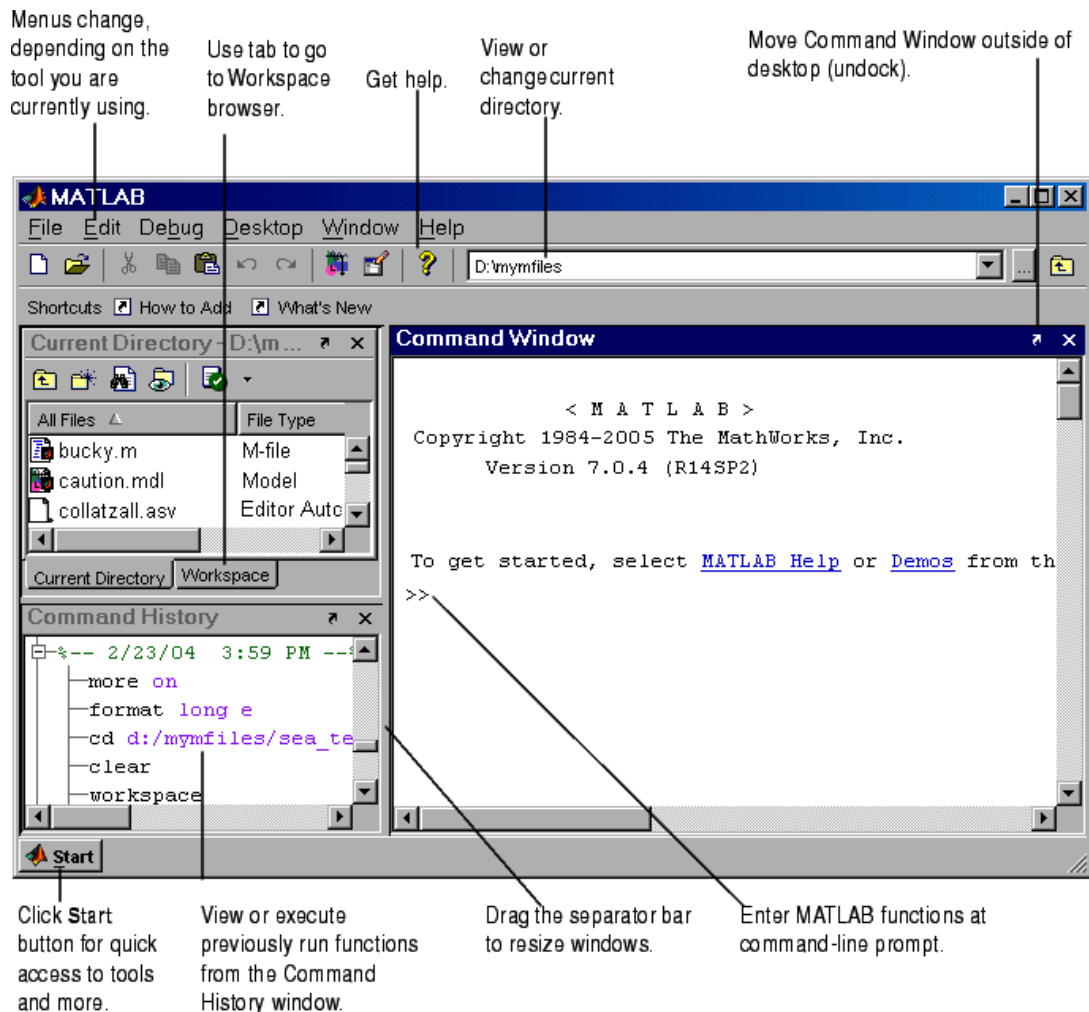


Figure 1.1: The graphical interface to the MATLAB workspace

When MATLAB is started for the first time, the screen looks like the one that shown in the Figure 1.1. This illustration also shows the default configuration of the MATLAB desktop. You can customize the arrangement of tools and documents to suit your needs.

Now, we are interested in doing some simple calculations. We will assume that you have sufficient understanding of your computer under which MATLAB is being run.

You are now faced with the MATLAB desktop on your computer, which contains the prompt (>>) in the Command Window. Usually, there are 2 types of prompt:

>>	for full version
EDU>	for educational version

NOTE: To simplify the notation, we will use this prompt, >>, as a standard prompt sign, though our MATLAB version is for educational purpose.

Using MATLAB as a calculator

As an example of a simple interactive calculation, just type the expression you want to evaluate. Let's start at the very beginning. For example, let's suppose you want to calculate the expression, $1 + 2 \times 3$. You type it at the prompt command (`>>`) as follows,

```
>> 1+2*3
ans =
    7
```

You will have noticed that if you do not specify an output variable, MATLAB uses a default variable `ans`, short for *answer*, to store the results of the current calculation. Note that the variable `ans` is created (or overwritten, if it is already existed). To avoid this, you may assign a value to a variable or output argument name. For example,

```
>> x = 1+2*3
x
=
    7
```

Table 1.1: Basic arithmetic operators

SYMBOL	OPERATION	EXAMPLE
+	Addition	$2 + 3$
−	Subtraction	$2 - 3$
*	Multiplication	$2 * 3$
/	Division	$2 / 3$

Creating MATLAB variables

MATLAB variables are created with an assignment statement. The syntax of variable assignment is

variable name = a value (or an expression)

For example,

```
>> x = expression
```

where `expression` is a combination of numerical values, mathematical operators, variables, and function calls. In other words, `expression` can involve:

- manual entry
- built-in functions
- user-defined functions

1.4.2 Overwriting variable

Once a variable has been created, it can be reassigned. In addition, if you do not wish to see the intermediate results, you can suppress the numerical output by putting a semicolon (;) at the end of the line. Then the sequence of commands looks like this:

```
>> t = 5;
>> t = t+1
t
    =
     6
```

Error messages

If we enter an expression incorrectly, MATLAB will return an error message. For example, in the following, we left out the multiplication sign, *, in the following expression

```
>> x = 10;
>> 5x
??? 5x
    |
Error: Unexpected MATLAB expression.
```

Controlling the hierarchy of operations or precedence

Let's consider the previous arithmetic operation, but now we will include *parentheses*. For example, $1 + 2 \times 3$ will become $(1 + 2) \times 3$

```
>> (1+2)*3
Ans
    = 9
```

and, from previous example

```
>> 1+2*3
Ans
    = 7
```

By adding parentheses, these two expressions give different results: 9 and 7.

The order in which MATLAB performs arithmetic operations is exactly that taught in high school algebra courses. *Exponentiations* are done *first*, followed by *multiplications* and *divisions*, and finally by *additions* and *subtractions*. However, the standard order of precedence of arithmetic operations can be changed by inserting *parentheses*. For example, the result of $1+2 \times 3$ is quite different than the similar expression with parentheses $(1+2) \times 3$. The results are 7 and 9 respectively. Parentheses can always be used to overrule *priority*, and their use is recommended in some complex expressions to avoid ambiguity.

Therefore, to make the evaluation of expressions unambiguous, MATLAB has estab-

lished a series of rules. The order in which the arithmetic operations are evaluated is given in Table 1.2. MATLAB arithmetic operators obey the same *precedence* rules as those in

Table 1.2: Hierarchy of arithmetic operations

PRECEDENCE	MATHEMATICAL OPERATIONS
First	The contents of all parentheses are evaluated first, starting from the innermost parentheses and working outward.
Second	All exponentials are evaluated, working from left to
right Third	All multiplications and divisions are evaluated,
working	from left to right
Fourth	All additions and subtractions are evaluated, starting from left to right

most computer programs. For operators of *equal* precedence, evaluation is from *left to right*. Now, consider another example:

$$\frac{1}{2+3^2} + \frac{4}{5} \times \frac{6}{7}$$

In MATLAB, it becomes

```
>> 1/(2+3^2)+4/5*6/7
ans = 0.7766
```

or, if parentheses are missing,

```
>> 1/2+3^2+4/5*6/7
ans = 10.1857
```

Entering multiple statements per line

It is possible to enter multiple statements per line. Use commas (,) or semicolons (;) to enter more than one statement at once. Commas (,) allow multiple statements per line without suppressing output.

```
>> a=7; b=cos(a), c=cosh(a)
b =
    0.6570
c =
   548.3170
```

Miscellaneous commands

Here are few additional useful commands:

- To clear the Command Window, type `clc`
- To abort a MATLAB computation, type `ctrl-c`

- To continue a line, type ...

Quitting MATLAB

To end your MATLAB session, type **quit** in the Command Window, or select **File→ Exit MATLAB** in the desktop main menu.

Getting help

To view the online documentation, select [MATLAB Help](#) from Help menu or [MATLAB Help](#) directly in the Command Window. The preferred method is to use the *Help Browser*. The Help Browser can be started by selecting the ? icon from the desktop toolbar. On the other hand, information about any command is available by typing

```
>> help Command
```

Another way to get help is to use the `lookfor` command. The `lookfor` command differs from the `help` command. The `help` command searches for an exact function name match, while the `lookfor` command searches the quick summary information in each function for a match. For example, suppose that we were looking for a function to take *the inverse of a matrix*. Since MATLAB does not have a function named `inverse`, the command `help inverse` will produce nothing. On the other hand, the command `lookfor inverse` will produce detailed information, which includes the function of interest, `inv`.

```
>> lookfor inverse
```

NOTE - At this particular time of our study, it is important to emphasize one main point. Because MATLAB is a huge program; it is impossible to cover *all the details* of each function one by one. However, we will give you information how to get help. Here are some examples:

- Use on-line `help` to request info on a specific function

```
>> help sqrt
```

- In the current version (MATLAB version 7), the `doc` function opens the on-line version of the help manual. This is very helpful for more complex commands.

```
>> doc plot
```

- Use `lookfor` to find functions by keywords. The general form is

```
>> lookfor FunctionName
```